

Learning that can show its work.

A private local learning workspace connecting source evidence, editable concept experiments and fresh transfer practice.

The screenshot displays the TraceLearn interface, which is divided into three main sections:

- Left Sidebar (Navigation):**
 - TraceLearn** logo and name.
 - YOUR WORKSPACE:** A dropdown menu currently set to "Database foundations". Below it is a link to "Import your own notes".
 - LEARNING PATH:** A list of 10 topics:
 - 01 SQL NULL (highlighted with a right arrow)
 - 02 Aggregates
 - 03 NOT IN
 - 04 Candidate keys
 - 05 Functional dependencies
 - 06 Attribute closure
 - 07 Second normal form
 - 08 3NF vs BCNF
 - 09 Decomposition
 - 10 LEFT JOIN
- Main Content Area:**
 - At the top, a question: "For salaries 100, 200, and NULL, which rows survive WHERE salary <> 100, and why?" with a progress indicator "81/1200" and an "Ask the source" button.
 - Below the question, a status bar shows "Automated support checks passed" and "16.1s · qwen3.5-9b".
 - CONCLUSION:** A text block stating: "For salaries 100, 200, and NULL, the WHERE clause salary <> 100 returns only the row with 200 because the comparison for 100 yields FALSE and the comparison for NULL yields UNKNOWN, both of which are filtered out."
 - Two source excerpts are provided:
 - Source 1 (null-where):** "SQL uses TRUE, FALSE, and UNKNOWN in its three-valued logic. A WHERE clause keeps a row only when its predicate is TRUE; FALSE and UNKNOWN are both filtered out. For salaries 100, 200, and NULL, WHERE salary <> 100 returns only the row with 200. To include missing salaries, write salary <> 100 OR salary IS NULL."
 - Source 2 (null-comparison):** "NULL represents missing or unknown information. It is not zero, an empty string, or a value equal to another NULL. Ordinary SQL comparisons involving NULL, including NULL = NULL and NULL <> 5, evaluate to UNKNOWN rather than TRUE or FALSE. Use IS NULL or IS NOT NULL to test
- Right Sidebar (Source Evidence):**
 - Section: "SQL USES TRUE, FALSE, and UNKNOWN in its three-valued logic. A WHERE clause keeps a row only when its predicate is TRUE; FALSE and UNKNOWN are both filtered out. For salaries 100, 200, and NULL, WHERE salary <> 100 returns only the row with 200. To include missing salaries, write salary <> 100 OR salary IS NULL."
 - SQL - THREE-VALUED LOGIC:** A table showing the result of the query "salary <> 100".

salary	predicate	WHERE
100	FALSE	excluded
200	TRUE	✓ kept
NULL	UNKNOWN	excluded
 - Section: "null-where" with a note: "Source text is inspectable. AI interpretations still need judgment."
 - Browse passages:** A dropdown menu showing "p.1 · WHERE keeps only TRUE".
 - Source fingerprint:** A section for tracking the source of the information.

Version 1.3.0. The actual application displays one accepted conclusion with up to two exact source excerpts. This screenshot is scripted software acceptance, not a study participant.

This report describes a working application and author-created technical evaluations. It makes no claim of improved grades, validated mastery, independent human assessment, user adoption or competition ranking.

Source and downloads: github.com/FENGJIA666/tracelearn / v1.3.0

1. A learning loop, not just an explanation

Fluent text can feel familiar without demonstrating that a learner can use a concept. TraceLearn asks a prediction, grades against an original fixed answer key, links the relevant source passage, and offers a fresh transfer question. A wrong option suggests a possible misconception; it does not diagnose a person or establish durable learning.

The first scenario is SQL NULL: predict which rows survive value ≤ 100 , inspect TRUE/FALSE/UNKNOWN, change the condition to include missing values, then answer a related question independently. The Counterexample Lab also computes attribute closure and checks candidate-key minimality under explicitly shown dependencies.


TraceLearn.

YOUR WORKSPACE

Database foundations ▾

+ Import your own notes ↗

LEARNING PATH 10

01 SQL NULL >

02 Aggregates

03 NOT IN

04 Candidate keys

05 Functional dependencies

06 Attribute closure

07 Second normal form

08 3NF vs BCNF

09 Decomposition

10 LEFT JOIN

 Learning notebook 2



Your notes. Your device.

Runs locally after model download.
No API key required.

[Local setup](#) ↗

BUILD CHALLENGE 3.0 2026

☒ Candidate key: determines everything; no removable attribute

☐ Superkey, but not a candidate key: an attribute is removable

☐ Not a superkey: some attributes cannot be derived

 **The computation reveals a different result.**

$\{A, D, B\}^+ = \{A, B, C, D\}$

- Start with the selected attributes $\{A, D, B\}$
- $B \rightarrow C$ adds $\{C\}$
→ $\{A, B, C, D\}$
- No rule can add another attribute. Stop.

This is a superkey, but not a candidate key. The closure covers the complete relation.

Try removing each attribute

Remove	Closure without it	Still a superkey?
A	$\{B, C, D\}$	No — needed
D	$\{A, B, C\}$	No — needed
B	$\{A, B, C, D\}$	Yes — removable

Closure is monotone: if removing any single attribute loses the superkey property, no smaller subset can be a superkey.

 [Make another prediction](#)

☐ closure
 ☒ keys

Results follow the stated dependencies. A small data sample cannot establish these constraints, and this trace does not automatically prove every normal form.

Try a transfer question →

[Export lab record \(7\)](#)

Counterexample Lab from the previous verified release; the underlying SQL and closure computations are preserved in v1.3. Navigation now matches the selected concept and restores its source on refresh.

Three-minute judge route	Action
Predict	Choose an answer and confidence; open the referenced passage.
Change the conditions	Edit SQL values or key attributes; compare the computed trace with your prediction.
Transfer	Open the concept-matched fresh question, without revealing an earlier answer.
Reflect	Reopen saved local answers and their evidence; export quiz and AI history.

2. Current implementation

Layer	Implementation
Interface	React + TypeScript; English/Chinese controls; desktop and narrow layouts; cancellation and recovery.
Local service and storage	Node.js / Express on 127.0.0.1; SQLite source hashes, vectors, questions, attempts and chats.
Documents	Text PDF, UTF-8 Markdown and TXT; 10 MB per file. PDF: 50 pages; Markdown/TXT: 175,000 UTF-16 code units. Exact source IDs, locations and file SHA-256.
Retrieval	Top five passages ranked by 0.75 cosine similarity + 0.25 token overlap, with local embeddings.
Runtime models	qwen3.5:9b for answers and cloze distractors; qwen3-embedding:0.6b for embeddings. Ollama 0.34.0.
Answer configuration	Temperature 0, seed 42, context 16384, output cap 1800; think:false and truncate:false. Exact parameters and model digests accompany the source.
Practice	20 original passages, 10 diagnostic/transfer pairs; fixed grading. Imported-note practice uses source-derived cloze answers.

One-conclusion evidence pipeline

The server enumerates literal excerpts with original UTF-16 offsets. The model selects at most two excerpt IDs for one concise conclusion; it cannot write the quotation text. The server resolves IDs, checks schema and exact location, then makes a separate call to the same local model to assess answerability, every clause and source attribution, its own citations and coverage of applicable conditions. Only the accepted conclusion forms the final answer.

The app distinguishes an accepted answer, source insufficiency and an unverified response. A second draft/review attempt is allowed: at most four model calls and 180 seconds across retrieval and generation. Long input must fail explicitly instead of silently losing context. These controls improve inspectability; the same model can still make correlated drafting and review mistakes.

3. Source recall and local operation

For imported notes, a deterministic planner selects an unused source sentence and an exact phrase as the answer. The local model supplies three distractors; the keyed option must reconstruct the original sentence. Duplicate wording and the explicit SQL non-NULL alias family are rejected, with at most one dedicated quality repair after the two-call base generator. This remains verbatim source recall, not a validated semantic discrimination or transfer assessment.

The learning notebook reopens saved question/claim/citation records without rerunning inference. Older records retain their original quote-only check status. Exports preserve each claim-to-quote relationship and distinguish an unsuccessful check from a finding of source insufficiency.

Run it locally

```
npm run setup
npm start
http://127.0.0.1:4317
```

Node.js 22.13+ and Ollama 0.34.0+ are required; setup checks the Ollama minimum for tested chat truncate:false behavior. Initial downloads of the two application models total approximately 7.23 GB, plus packages. No API key, paid inference service, user account, CDN, external font or hosted model is required. Once installed, inference and study material remain on the computer. Exact tested model digests are checked by setup; weights are downloaded separately. Reproducing the frozen comparison additionally needs ollama pull qwen3:4b (about 2.5 GB), not required for normal application use.

The Mac launcher rebuilds this copy and compares installation/build identity before reusing a server. It selects another local port for a different copy or stale build. Portable practice separately provides the original course, deterministic experiments, recorded evidence and exports without installing Node.js or models; live AI and importing require the full application.

Measured runtime boundary

On Apple M4 Pro with 24 GB memory, a process policy denying outbound network except localhost allowed real inference. After restarting Ollama, the measured request took 25.79 s; the warm repeat took 16.00 s. The cold request used fresh isolated storage with no cached passage vectors; the warm request repeated it after the first completed request. Ollama reported 8.31 GB allocated across the loaded production models; this is a snapshot, not peak whole-system memory.

This verifies the app and model process boundary after installation; it does not claim that the whole computer or browser network was disconnected. macOS is the tested platform; Windows/Linux and direct file:// automation remain unverified. Portable static HTTP operation was checked.

4. New evaluation: freeze before opening the holdout

Version 1.3 adds 80 original cases based on four small fictional source documents: database rules, library policy, laboratory procedure and a shuttle timetable. Each 40-case split contains 16 source-answerable questions, 12 unanswerable questions and 12 contradicted-premise questions, with 20 English and 20 Chinese prompts. Developer and holdout questions share documents and themes; this is not generalization to unseen domains.

The dataset author prepared and locked both splits. The root implementation process did not open the sealed holdout before freezing the selected pipeline, model identities, explicit options, runner and scoring protocol. The runner rejects mismatched configurations before opening a holdout path. Agents share a filesystem, so this is procedural isolation rather than cryptographic blinding. The scoring agent also authored the dataset: review is not independent or blinded.

The comparator is the previous production grounded-answer configuration, using Qwen3-4B and its original quote validator. The current configuration changes the model and answer pipeline. Both use the same source material, questions and hybrid retriever. Results compare complete configurations; they do not isolate the effect of model weights or the review call. The older full-source-versus-retrieval experiment remains separate.

Strict task correctness requires a complete source-consistent answer and valid evidence, or a justified refusal under the saved protocol. No-answer errors and unverified outputs stay in the denominator. Correct numbers paired with unsupported claims or wrong citations do not count as fully correct. Runtime self-review labels are not the scoring authority.

Held-out results: 40 questions, 80 requests

Saved technical result	Previous configuration	Current configuration
Strict task correctness	18/40	34/40
Unsupported accepted outputs	4/25 confirmed; 1 unresolved	4/39 confirmed
Unresolved semantic judgments	1	0
No delivered answer or refusal	15/40	1/40
Citation validity among delivered answers	13/19	24/28
Median wall time	5.79 s	16.84 s
95th percentile wall time	7.65 s	35.17 s

Percentile uses the nearest-rank method. Times include retrieval and retries; model loading and other local work can affect latency. Unresolved judgments remain incorrect in strict totals; an unresolved unsupported-claim judgment is shown separately instead of counted as a confirmed error or as zero. Small samples do not establish educational effectiveness.

5. Inspect every accepted answer and failure

The screenshot shows the TraceLearn interface. On the left is a sidebar with 'YOUR WORKSPACE' (Database foundations) and 'LEARNING PATH' (10 steps, with '01 SQL NULL' selected). The main area is titled 'ORIGINAL COURSE · 10 CONCEPT PAIRS' and 'From “I think” to “I understand.”'. It includes a 'Show the quick tour' link and tabs for 'Practice', 'Ask the source', 'Lab', 'Evidence' (selected), and 'Notebook'. Below the tabs are buttons for 'Current configuration - new evaluation' and 'Original experiment - historical'. A section titled 'CASE-BY-CASE EVIDENCE' contains the instruction 'Inspect the answer, its evidence, and its score.' and a warning box: 'RECORDED EXPERIMENT - NOT LIVE MODEL OUTPUT. Scored by the Codex agent that authored the dataset. No independent human review or learning-effect study.' Below this are dropdowns for 'Dataset split' (Holdout) and 'Filter by the current configuration's saved review' (All cases). A note states: 'Holdout totals from saved scoring judgments; unchanged by the result filter.' At the bottom, a table compares configurations:

Frozen configuration	Current configuration
18 / 40	34 / 40
Strictly correct / total records	Strictly correct / total records
Not delivered: 15	Not delivered: 1

The recorded-evidence view displays the question, reference, both final outputs, actual selected quotations, full retrieved inputs and saved Codex-assisted scoring reasons. It is explicitly labeled as a recorded run, not live AI.

Development results: 40 questions, 80 requests

Saved technical result	Previous configuration	Current configuration
Strict task correctness	20/40	38/40
Unsupported accepted outputs	6/30 confirmed	1/39 confirmed
Unresolved semantic judgments	0	0
No delivered answer or refusal	10/40	1/40
Citation validity among delivered answers	17/22	27/28
Median wall time	4.46 s	13.51 s
95th percentile wall time	6.15 s	22.88 s

Development prefixes and targeted probes were used to change the implementation and select the local model. They remain in evidence/v1.3, including unsuccessful attempts. They are not new test sets or independent repeated trials. Only the recorded final development configuration and the one-shot held-out comparison appear in the tables above.

6. Verification and remaining limitations

Boundary	Current evidence
Automated checks	123 tests passed; 11 additional isolated HTTP checks. Final production and portable builds completed.
Actual browser	Actual isolated Chrome: both full configuration comparisons, own-source citations, filtering, keyboard focus, 390px English/Chinese, original-source navigation, portable practice/experiments/transfer, real downloads and refresh. In-app browser: real local AI, source jump, saved-answer refresh and export event; actual narrow answer layout.
Controlled UI	Earlier mocked API acceptance checked support/refusal/unverified/legacy rendering and cancellation. These checks are distinct from the real local model runs.
Local AI	Final r8: three real original-course answers, exact citations and preserved SQLite history; controlled cold/warm requests plus one actual UI request. r8 long-input/overflow checks are retained. The earlier r7 import/cloze/cancel fixture is separately labelled.
Network boundary	Both app and Ollama were run under a localhost-only process policy; external hostname and literal-IP curl probes failed. Real answers succeeded. Model-unavailable errors and recovery were observed. Whole-computer networking was unchanged.
Packaging	Fresh locked npm ci, final source/build equality, actual Mac launcher and served assets passed in an isolated installation. The package guard passed 16 synthetic checks. Final archive and anonymous downloads are checked separately after packaging.

The original SQL simulator has 1,600 row-result comparisons against SQLite. Candidate-key results were checked for every subset of both example schemas against a separately implemented finite-relation oracle. These are independent computational checks of bounded code, not independent human evaluation of learning.

What this evidence does not establish

Exact source selection prevents invented quote text but cannot prove an interpretation. Same-model review may accept an incorrect claim or reject a correct one. Retrieval can omit needed passages. Source recall can still contain semantically weak distractors beyond the explicit alias checks. Prompt-injection resistance is bounded by representative tests, not a universal guarantee.

OCR, arbitrary SQL execution, multi-user accounts and public inference hosting are outside scope. PDF reading order can be imperfect. Fixed practice correctness is not psychometric mastery. There were no consented student participants, delayed retention measurements or measured grade gains. No contest score or rank has been observed.

A preserved historical baseline

The original v1.0 experiment contains 80 cases and 160 requests comparing full-source input with hybrid retrieval under the same Qwen3-4B pipeline. Its data, original course and server/ai.ts remain unchanged. Versions 1.1/1.2 added portable access and deterministic experiments without rewriting those scores. Historical measurements are never presented as current-model evidence.

7. Contribution, licenses and sources

Codex substantially assisted with product design, implementation, original teaching and evaluation content, debugging, visual work, testing, release preparation and documentation. Local Qwen models produce runtime responses. Dataset author and semantic reviewer are the same Codex-assisted process; this is explicitly not independent human validation. The organizer has not provided separate approval of AI-assisted code authorship; no such permission is claimed.

Original application code and course materials are MIT-licensed. Qwen model weights use Apache-2.0 and are obtained separately. Exact locked dependency notices and model links appear in THIRD-PARTY-NOTICES.md. Study material is stored locally; no personal runtime database or user credentials are included in the package.

Reproduce and audit

evaluation-v13/ contains locked sources, cases and the scoring protocol. Each final run retains raw JSONL, HTTP model requests/responses, source snapshots, timings, model options and per-output judgments. The final manifest binds the files by SHA-256. Model-produced review rationales and Codex-assisted evaluation judgments are stored separately.

Primary technical references

Ollama model and model-weight license: <https://ollama.com/library/qwen3.5:9b>

Local embedding model: <https://ollama.com/library/qwen3-embedding:0.6b>

Ollama 0.34 request and truncation handling: <https://github.com/ollama/ollama/blob/v0.34.0/server/routes.go>

SQL comparison predicates: <https://www.postgresql.org/docs/current/functions-comparison.html>

SQL aggregate functions: <https://www.postgresql.org/docs/current/functions-aggregate.html>

SQL table expressions and outer joins: <https://www.postgresql.org/docs/current/queries-table-expressions.html>

Competition rules: <https://ml-build-challenge-3.devpost.com/rules>

Public submission: <https://devpost.com/software/tracelearn-learning-that-can-show-its-work>

Current provenance

The complete release links source code, a portable practice file, this report, model settings, raw evaluations and reproducible acceptance records. The submitted project is updated in place; final submission state is verified separately from local preparation.

Report generated from saved evidence on 16 September 2026, 01:45 SGT.